

When Should a Vision Specialist Defer?

Supplementary Material: Learned Routing

1 Experimental Scope and Naming

Learned Routing (LR) denotes the shared pre-call input and routing framework. The four-state instantiation is LR_{4S}; LR_{PH} adapts the post-hoc deferral loss [1] to the same backbone and visible features. The main experiment uses normalized LR_{4S}, abbreviated LR. These names distinguish our implementations, not newly invented versions of the underlying deferral loss.

The main experiment and Table 3 use different training settings. The main experiment uses 80 newly fitted router heads with full-budget validation selection. The training-objective and scoring comparison uses the earlier 0–25% checkpoint-selection rule for all trained variants. Its four four-state score rows share identical weights. We do not interpret its comparison with post-hoc deferral as a comparison with the final main results under identical training settings. Test sets have been inspected during method development, so this is an exploratory revision rather than an independent blind confirmation.

2 Data Preparation and Evaluation

Table 1 lists the examples actually used in this study. The 4 datasets span detection-related localization and event recognition, together with classification and visual reasoning. They are not 4 object-detection benchmarks, and their task scores have different meanings.

Table 1: Experimental splits. Counts refer to task examples, not uniformly to distinct images. Training denotes the router’s full available training split; specialist training provenance is listed separately. Totals are the sums of the three columns after the recorded exclusions.

Dataset	Example unit	Train	Validation	Test	Total	Evaluation
CUB-200-2011	One image	4,792	1,200	5,794	11,786	Top-1 accuracy
gRefCOCO	Image-expression pair	209,344	14,229	35,263	258,836	Strict localization accuracy
NLVR2	Two images and a statement	85,042	6,982	13,937	105,961	Binary accuracy
ConstructionSite10k	One image, 4 event labels	6,308	701	3,004	10,013	Event macro-F1

CUB-200-2011 [2] tests fine-grained recognition among 200 bird species, where differences in plumage, shape and local appearance distinguish related categories. We report top-1 class accuracy. The router reserves 1,200 images from the official training partition for validation and excludes two training images identified as near-duplicates of test images, leaving 4,792 training examples. The official 5,794-image test partition is retained.

gRefCOCO [3] tests generalized referring-expression localization, including expressions with one, multiple or no matching objects. TestA and testB contribute 19,200 and 16,063 expressions. Our fixed box-based evaluation requires a valid empty answer for no-target examples. Boxes from either model are first filtered at score ≥ 0.7 . For target-present examples, greedy one-to-one matching uses generalized intersection over union of at least 0.5; an example is correct only when instance F1 equals one, with no missed or extra instances. This strict correctness measure is distinct from the dataset’s official segmentation evaluation and from detection average precision.

NLVR2 [4] tests whether a natural-language statement is true of an ordered pair of natural images, including counting, attributes and relations across the pair. We report binary accuracy. The prepared training set contains 85,042 usable examples after filtering the source training partition; validation has 6,982, and the two test portions have 6,967 and 6,970. Routing acts on the complete two-image example.

ConstructionSite10k [5] contains construction scenes with structured safety annotations. We evaluate 4 event-presence categories: personal-protection, fall-protection, unprotected-edge and excavator-proximity violations. The original 7,009 training images are split into 6,308 for training and 701 for validation; all 3,004 test images are retained. The dataset name is approximate: the prepared partitions contain 10,013 images. We report

$$M_{\text{event}} = \frac{1}{4} \sum_{j=1}^4 \frac{2TP_j}{2TP_j + FP_j + FN_j}, \quad (1)$$

with zero for a class whose denominator is zero. Delegation replaces all 4 event predictions for an image; confusion counts are then recomputed. This is event recognition, not bounding-box average precision.

2.1 Examples from the 4 Datasets

CUB-200-2011



Nashville Warbler (class 172)

gRefCOCO



Query: a mouse beside a laptop and a little boy sitting far from us is looking at a laptop.
Targets: 2.

NLVR2



Statement: The right image shows a curving walkway of dark glass circles embedded in dirt and flanked by foliage. Label: False.

ConstructionSite



4 safety-event labels: all absent.

Figure 1: Real validation examples and their recorded annotations. CUB and ConstructionSite use one image, NLVR2 uses an ordered image pair, and gRefCOCO couples an image with an expression. The first record in each validation manifest was selected before viewing the images. Images illustrate the tasks rather than model performance.

The gRefCOCO example refers to two separate targets, illustrating why all matching objects must be found. The NLVR2 label evaluates the full statement about the ordered pair. For ConstructionSite, an all-negative image still requires evaluating all 4 event categories. The sample identifiers and original annotations are provided with the figure data.

3 Full-Budget Training and Threshold Selection

The specialist and 8.07M-parameter distilled image/text backbone are frozen. Image features, task text where present, hidden states at 90% specialist depth, prediction summaries and confidence enter separate 128-dimensional projections and layer normalization. Their concatenation feeds a 128-hidden-unit GELU MLP. Training uses paired cross-entropy, uniform sampling, AdamW at 0.003, weight decay 0.01, batch size 256, at most 50 epochs and patience 8. Seeds are 2026–2030. Training inputs alone determine feature standardization.

With paired state probabilities p_R, p_H, p_B , the score is

$$s_\alpha = \frac{p_R - p_H}{(p_R + p_H + 2p_B + 10^{-8})^\alpha}, \quad \alpha \in \{0, 0.25, 0.5, 0.75, 1\}.$$

For every epoch and candidate α , we sort validation inputs and evaluate exactly $\lfloor kn/100 \rfloor$ calls, $k = 0, \dots, 100$, retaining original order at ties. Trapezoidal area over actual call fraction selects the epoch and α . A tie within an epoch favors smaller α ; an epoch tie retains the earlier checkpoint. Early stopping uses this criterion. Cross-entropy itself is unchanged. Historical runs saved only their selected checkpoints, so changing checkpoint selection required refitting the 80 heads.

After weights and α are fixed, a separate validation search selects one threshold. At each interior budget, the threshold is the score of the last allowed input. All threshold ties are included only if they fit the quota; otherwise all equal-score ties are excluded. The highest validation metric wins; ties favor fewer calls, then a smaller budget. Budgets 0 and 100% explicitly mean never and always call. Confidence uses the same endpoint-inclusive search. Test labels do not select weights, normalization or thresholds. Actual test call rates can differ after a threshold is transferred.

Table 2: Selected normalization strength α for the main experiment, separately chosen on validation data for each pair and seed. M: MiniCPM. Q: Qwen. Across 80 runs, 0, 0.25, 0.5, 0.75, 1 are selected 29, 8, 9, 9 and 25 times. These choices do not describe the earlier Table 3 checkpoints.

Task	Specialist	VLM	2026	2027	2028	2029	2030
CUB	ResNet-50	M	0	0	0.25	0	0
CUB	ResNet-50	Q	0	0	0	0	0.25
CUB	GLSim	M	0.5	0.25	0.5	0.5	0.5
CUB	GLSim	Q	0	0	0	0	0
gRefCOCO	GroundingDINO	M	0	0	0	0	0
gRefCOCO	GroundingDINO	Q	0	0	0	0	0
gRefCOCO	InstanceVG	M	0.75	0.75	0.75	0.75	0.75
gRefCOCO	InstanceVG	Q	1	1	1	1	1
NLVR2	BEiT3	M	1	1	1	1	1
NLVR2	BEiT3	Q	1	1	1	1	1
NLVR2	ViLT	M	1	1	1	1	1
NLVR2	ViLT	Q	1	1	1	1	1
Construction	YOLO26X	M	0.5	0.5	0.75	0.75	0.5
Construction	YOLO26X	Q	0.75	0.5	0.25	0.25	0.75
Construction	RT-DETR-X	M	0	0	0.25	0	0
Construction	RT-DETR-X	Q	0	0.25	0.25	0	0.5

4 Main Results and Full Curves

Table 3 includes both splits. Numerical per-seed cutoffs, comparison operators, chosen epochs and normalization strengths are stored with the revision data. Higher routed performance and fewer calls are different objectives; selected points need not dominate a baseline on both. Zero- and full-call endpoints agree with direct standalone evaluation.

Table 3: LR results are means over five router seeds (2026–2030); the supplement reports standard deviations. Complete four-way comparison. Scores and actual call rates are percentages. Thresholds selected on the 0–100% validation grid transfer unchanged to test. Parentheses give LR-minus-confidence differences in percentage points before rounding. Green marks improvement and red degradation: higher scores and fewer calls are better. Bold marks the higher routed score; † exceeds both test endpoints. M: MiniCPM-V-4.5; Q: Qwen3.8-27B-FP8.

Task	Specialist	VLM	Validation						Test					
			spec.	vlm.	conf.	call _{conf}	LR↑	call _{LR} ↓	spec.	vlm.	conf.	call _{conf}	LR↑	call _{LR} ↓
CUB-200-2011	ResNet	M	98.58	27.75	98.58	0.00	98.62 (↑ 0.03)	0.40 (↑ 0.40)	81.43	26.56	81.43	0.00	81.43 (=0.00)	0.59 (↑ 0.59)
	ResNet	Q	98.58	55.08	98.58	0.00	98.58 (=0.00)	0.00 (=0.00)	81.43	54.82	81.43	0.00	81.43 (=0.00)	0.00 (=0.00)
	GLSim	M	98.92	27.75	98.92	0.00	98.95 (↑ 0.03)	0.40 (↑ 0.40)	90.16	26.56	90.16	0.00	90.13 (↓ 0.03)	0.85 (↑ 0.85)
	GLSim	Q	98.92	55.08	98.92	0.00	98.92 (=0.00)	0.00 (=0.00)	90.16	54.82	90.16	0.00	90.16 (=0.00)	0.00 (=0.00)
gRefCOCO	GroundingDINO	M	56.05	64.12	64.12	100.00	65.91 (↑ 1.79)	45.80 (↓ 54.20)	26.69	50.01	50.01	100.00	50.23 [†] (↑ 0.22)	58.19 (↓ 41.81)
	GroundingDINO	Q	56.05	80.60	80.60	100.00	81.41 (↑ 0.80)	77.60 (↓ 22.40)	26.69	74.47	74.47	100.00	75.12 [†] (↑ 0.65)	88.47 (↓ 11.53)
	InstanceVG	M	75.03	64.12	77.17	64.99	75.67 (↓ 1.50)	14.40 (↓ 50.60)	66.95	50.01	61.88	52.09	67.91 [†] (↑ 6.04)	19.04 (↓ 33.06)
	InstanceVG	Q	75.03	80.60	82.02	71.00	83.38 (↑ 1.36)	45.60 (↓ 25.40)	66.95	74.47	72.43	68.39	76.24 [†] (↑ 3.81)	62.00 (↓ 6.38)
NLVR2	BEiT3	M	79.82	89.90	90.05	85.88	89.99 (↓ 0.05)	91.39 (↑ 5.51)	80.00	90.74	90.77	85.80	90.80 [†] (↑ 0.03)	91.10 (↑ 5.30)
	BEiT3	Q	79.82	90.50	90.58	85.88	91.09 (↑ 0.51)	84.19 (↓ 1.69)	80.00	90.06	90.24	85.80	90.60 [†] (↑ 0.36)	84.34 (↓ 1.46)
	ViLT	M	75.57	89.90	89.97	95.99	89.93 (↓ 0.04)	98.20 (↑ 2.21)	76.42	90.74	90.64	95.93	90.69 (↑ 0.05)	98.06 (↑ 2.13)
	ViLT	Q	75.57	90.50	90.59	95.99	90.50 (↓ 0.09)	99.60 (↑ 3.61)	76.42	90.06	89.96	95.93	90.07 [†] (↑ 0.10)	99.62 (↑ 3.69)
ConstructionSite	YOLO26x	M	18.25	20.67	22.64	93.87	28.71 (↑ 6.06)	45.73 (↓ 48.13)	18.13	22.36	25.23	92.54	25.88 [†] (↑ 0.65)	45.32 (↓ 47.22)
	YOLO26x	Q	18.25	25.45	26.43	92.87	32.80 (↑ 6.37)	45.14 (↓ 47.73)	18.13	27.00	25.97	92.21	29.80 [†] (↑ 3.82)	44.77 (↓ 47.44)
	RT-DETR-X	M	7.64	20.67	20.67	100.00	21.99 (↑ 1.33)	77.29 (↓ 22.71)	7.89	22.36	22.36	100.00	22.80 [†] (↑ 0.44)	78.44 (↓ 21.56)
	RT-DETR-X	Q	7.64	25.45	25.45	100.00	27.23 (↑ 1.78)	83.08 (↓ 16.92)	7.89	27.00	27.00	100.00	26.78 (↓ 0.22)	83.57 (↓ 16.43)

Table 4: Confidence cutoffs and mean LR validation budgets across five seeds. Each LR seed uses its own validation cutoff unchanged on test; exact per-seed cutoffs are provided in the accompanying data/per_seed.csv.

Task	Specialist	VLM	Conf. budget	Conf. cutoff	LR budget	LR cutoff
CUB-200-2011	ResNet	M	0	Never	0.4	Per seed
	ResNet	Q	0	Never	0.0	Per seed
	GLSim	M	0	Never	0.4	Per seed
	GLSim	Q	0	Never	0.0	Per seed
gRefCOCO	GroundingDINO	M	100	Always	45.8	Per seed
	GroundingDINO	Q	100	Always	77.6	Per seed
	InstanceVG	M	65	≥ −0.987771	14.4	Per seed
	InstanceVG	Q	71	≥ −0.994979	45.6	Per seed
NLVR2	BEiT3	M	86	> −0.999739	91.4	Per seed
	BEiT3	Q	86	> −0.999739	84.2	Per seed
	ViLT	M	96	≥ −0.99984	98.2	Per seed
	ViLT	Q	96	≥ −0.99984	99.6	Per seed
ConstructionSite	YOLO26x	M	94	≥ −0.490234	45.8	Per seed
	YOLO26x	Q	93	≥ −0.453369	45.2	Per seed
	RT-DETR-X	M	100	Always	77.4	Per seed
	RT-DETR-X	Q	100	Always	83.2	Per seed

Table 5: Selected outcome counts and gains are five-seed means; available outcome counts are fixed. Test outcome counts at the LR operating point. BC/BW: both correct/wrong; R/H: rescue/harm. Selected rescues, selected harms, missed rescues and avoided harms are counted separately. Construction counts 12,016 events, with routing over 3,004 images. Metric gains are recomputed; macro-F1 is not an event-count ratio.

Task	Specialist	VLM	BC	R	H	BW	Sel. R	Sel. H	Miss R	Avoid H	Gain (pp)
CUB-200-2011	ResNet	M	1,307	232	3,411	844	4.6	4.6	227.4	3,406.4	+0.000
	ResNet	Q	2,811	365	1,907	711	0.0	0.0	365.0	1,907.0	+0.000
	GLSim	M	1,433	106	3,791	464	1.0	2.8	105.0	3,788.2	-0.031
	GLSim	Q	3,032	144	2,192	426	0.0	0.0	144.0	2,192.0	+0.000
gRefCOCO	GroundingDINO	M	6,911	10,723	2,500	15,129	9,475.4	1,174.4	1,247.6	1,325.6	+23.540
	GroundingDINO	Q	7,441	18,820	1,970	7,032	18,720.8	1,641.6	99.2	328.4	+48.434
	InstanceVG	M	15,007	2,627	8,602	9,027	727.6	388.2	1,899.4	8,213.8	+0.962
	InstanceVG	Q	20,838	5,423	2,771	6,231	4,753.6	1,476.8	669.4	1,294.2	+9.292
NLVR2	BEiT3	M	10,421	2,225	729	562	2,189.4	684.6	35.6	44.4	+10.797
	BEiT3	Q	10,305	2,247	845	540	2,163.4	686.4	83.6	158.6	+10.598
	ViLT	M	9,940	2,706	711	580	2,693.8	704.8	12.2	6.2	+14.271
	ViLT	Q	9,761	2,791	890	495	2,788.2	886.8	2.8	3.2	+13.643
ConstructionSite	YOLO26x	M	8,176	1,198	2,109	533	1,139.6	803.4	58.4	1,305.6	+7.751
	YOLO26x	Q	9,508	1,282	777	449	1,231.6	315.0	50.4	462.0	+11.667
	RT-DETR-X	M	6,060	3,314	1,945	697	3,288.6	1,452.4	25.4	492.6	+14.912
	RT-DETR-X	Q	7,329	3,461	676	550	3,448.0	574.8	13.0	101.2	+18.896

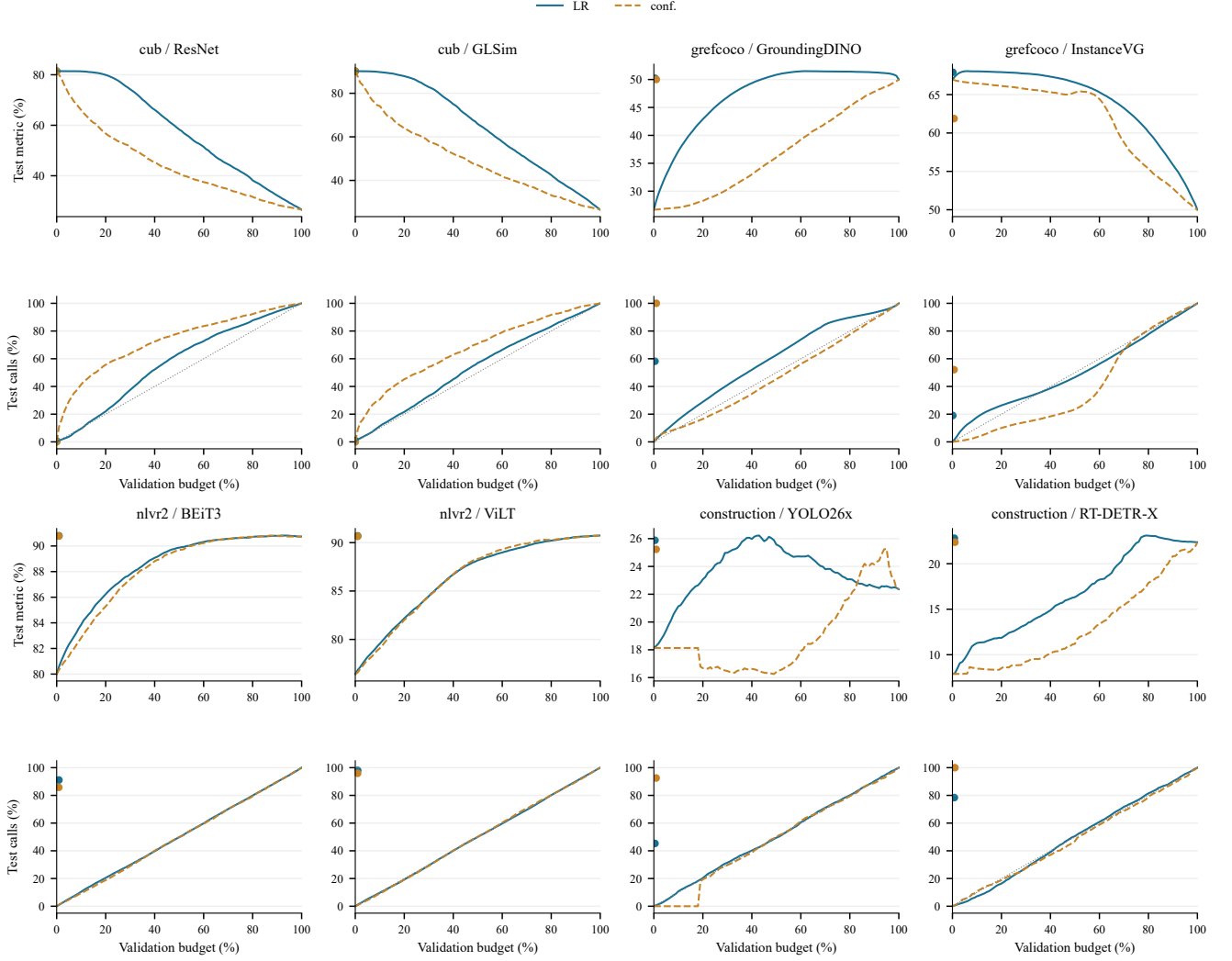


Figure 2: LR and confidence results for MiniCPM. Horizontal axes show the nominal validation budget; panels report test performance and transferred test call rates, including both endpoints.

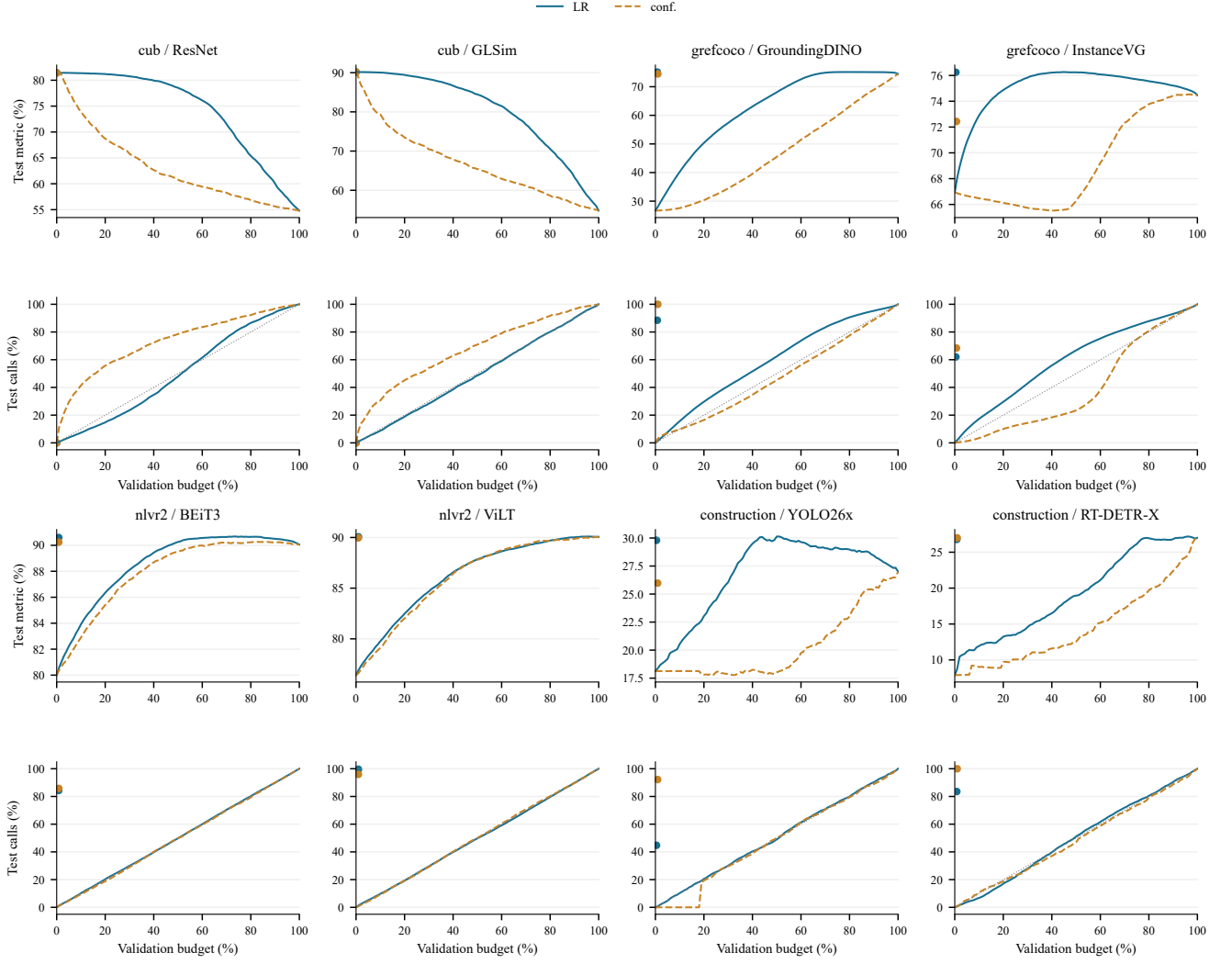


Figure 3: LR and confidence results for Qwen. Same conventions.

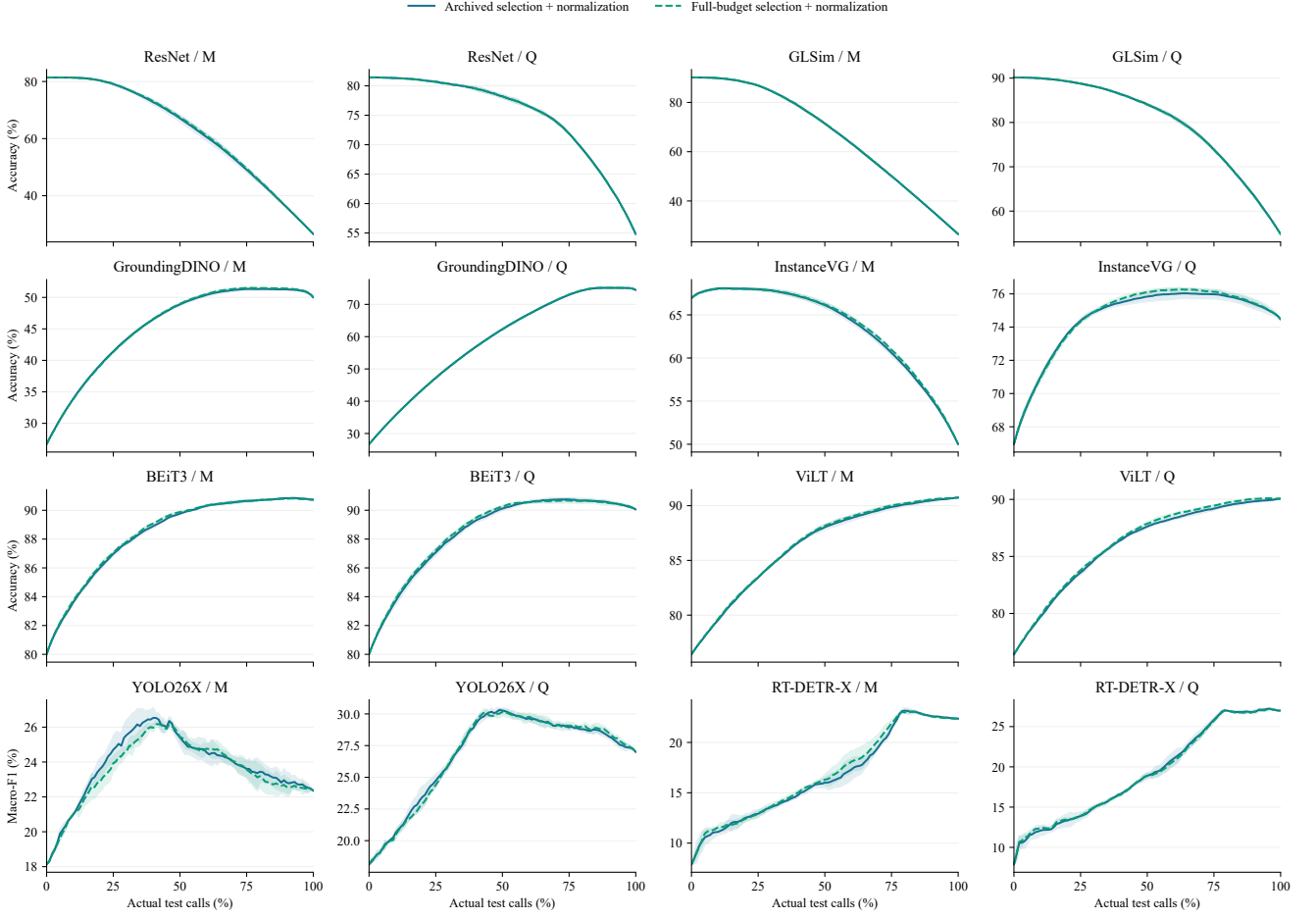


Figure 4: Old checkpoints with normalization versus full-budget checkpoint/normalization selection at identical actual calls. Five-seed means and standard-deviation bands. This isolates the incremental reselection effect after changing the score.

5 Training Objectives and Ranking Scores

This section gives the settings behind Table 3 of the main paper. The table evaluates the full 0–100% call curve, but the trained heads were selected using the earlier 0–25% validation criterion. Evaluation range and training selection range are distinct. In contrast, the main table selects weights using the full-budget criterion. Table 3 therefore compares objectives within the earlier experiments; it should not be read as a comparison of every objective under the final main-table training settings.

Four-state, three-state, gain and post-hoc runs use the same inputs, architecture except necessary head dimensions, optimizer settings, seeds and original 0–25% validation checkpoint-selection range. All four-state score variants share 80 identical checkpoints. Normalization strength alone is chosen by full-budget validation area.

Specialist and VLM answers remain fixed. Other archived post-hoc cost coefficients are not selected using test results. The reported averages are descriptive, not significance tests.

5.1 Targets, Outputs and All Seven Table Rows

Let $C = (1, 1)$, $R = (0, 1)$, $H = (1, 0)$ and $B = (0, 0)$ denote both-correct, rescue, harm and both-wrong outcomes, respectively, with specialist correctness first. A four-output head produces logits ℓ_k and probabilities $p_k = \exp(\ell_k) / \sum_j \exp(\ell_j)$ for $k, j \in \{C, R, H, B\}$. Thus p_R estimates the chance that replacement fixes an error, p_H that it causes an error, and p_B that both answers are wrong. For construction, the target q_k is the fraction of four event labels in state k . There is one predicted distribution and one score per image, and a call replaces all four event answers. Every row sorts scores in descending order: larger scores receive calls first.

Row 1: post-hoc deferral, score r . One unrestricted scalar output $r = f_\theta(\mathcal{I})$ replaces the four logits. With $e_S = 1 - c_S$ and $e_L = 1 - c_L$, training minimizes the sample mean of

$$e_S \log(1 + \exp(-r)) + e_L \log(1 + \exp(r)).$$

For construction, the two errors are averaged over events. A specialist error favors a positive score, and a VLM error a negative score. Added call cost is zero. Ranking uses r directly, without a probability conversion or a fixed zero threshold.

Row 2: three-state classification, score $p_R - p_H$. The targets are rescue, harm and unchanged ($U = C \cup B$). Three logits feed a three-way softmax. Training minimizes cross-entropy $-\sum_{k \in \{R, H, U\}} q_k \log p_k$, averaged over examples. Construction again uses event fractions as targets. The score subtracts predicted harm probability from predicted rescue probability. Both unchanged outcomes have zero gain and share one target.

Row 3: gain regression, score $\hat{\Delta}$. One unrestricted scalar $\hat{\Delta} = f_\theta(\mathcal{I})$ predicts $d = c_L - c_S$, which is +1 for rescue, −1 for harm and 0 for either unchanged outcome. Construction uses $d = \frac{1}{4} \sum_{j=1}^4 (c_{L,j} - c_{S,j})$. Training minimizes the sample mean of $(\hat{\Delta} - d)^2$. The scalar prediction is the ranking score, without softmax, sigmoid or clipping.

Row 4: four-state classification, probability difference. The four-state head minimizes cross-entropy $-\sum_{k \in \{C, R, H, B\}} q_k \log p_k$. Its score $p_R - p_H$ estimates expected correctness gain from replacement and lies in $[-1, 1]$. Neither unchanged outcome contributes to the difference. Rows 5–7 reuse exactly these head weights and probabilities.

Row 5: four-state classification, rescue-to-harm ratio. The score $\log(p_R/p_H) = \ell_R - \ell_H$ measures rescue relative to harm, instead of subtracting probabilities. The implementation subtracts logits, avoiding division by very small probabilities. The unchanged states do not enter the ratio explicitly.

Row 6: four-state classification, error ratio r_{4S} . The predicted specialist error rate is $p_R + p_B$, and the predicted VLM error rate is $p_H + p_B$. The score is

$$r_{4S} = \log \frac{p_R + p_B}{p_H + p_B} = \text{LSE}(\ell_R, \ell_B) - \text{LSE}(\ell_H, \ell_B),$$

where $\text{LSE}(a, b) = \log(\exp(a) + \exp(b))$ is evaluated by a numerically stable log-sum-exp operation. Both-wrong probability enters both error rates. For positive conditional error rates, setting the derivative of the expected row-1 loss to zero gives $\exp(r) = \mathbb{E}[e_S | \mathcal{I}] / \mathbb{E}[e_L | \mathcal{I}]$. Row 6 estimates this ratio from four-state probabilities, while row 1 fits a scalar separately. Their fitted rankings need not coincide.

Row 7: four-state classification, normalized gain s_α . The score is

$$s_\alpha = \frac{p_R - p_H}{(p_R + p_H + 2p_B + 10^{-8})^\alpha}.$$

The denominator is the sum of the two predicted error rates, with 10^{-8} added for numerical stability. At $\alpha = 0$, the score equals row 4. Ignoring that constant, at $\alpha = 1$ it is $(a - b)/(a + b)$ for $a = p_R + p_B$ and $b = p_H + p_B$, which orders examples like a/b and hence row 6. Intermediate values partially normalize gain. For Table 3, each pair and seed selects $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ by full-budget validation area with the row-4 checkpoint fixed. This differs from the main experiment’s joint epoch and α selection.

5.2 How the Table Values Are Calculated

For each variant, model pair and seed, sort the n test examples by score, preserving original order at ties. At budget index $k = 0, \dots, 100$, call the VLM on the first $m_k = \lfloor kn/100 \rfloor$ examples and retain the specialist elsewhere. Let $b_k = m_k/n$ and M_k be the resulting task metric as a fraction. The full-budget mean in percentage units is

$$A = 100 \sum_{k=1}^{100} \frac{M_{k-1} + M_k}{2} (b_k - b_{k-1}).$$

The interval runs from $b_0 = 0$ to $b_{100} = 1$, so no further range normalization is needed. This is a trapezoidal average of the entire curve, not the best test threshold. For CUB, gRefCOCO and NLVR2, M_k is task accuracy. For construction, the combined predictions determine new true-positive, false-positive and false-negative counts at each budget. Then M_k averages $2\text{TP}/(2\text{TP} + \text{FP} + \text{FN})$ over four events, assigning zero when the denominator is zero.

Each task column averages A over four model pairs and five seeds. For each seed, Avg. averages four pairs within each task and then gives equal weight to the four tasks. The displayed Avg. is the mean of those five seed-level values, and its SD uses divisor $5 - 1$. The final column subtracts the unrounded post-hoc Avg. from the unrounded variant Avg. and rounds to two decimals. Subtracting displayed rounded means can therefore differ by 0.01 points. No value pools examples across tasks or selects the best seed.

Table 6: Full-budget means (%). Avg.: equal-task mean (five-seed SD). Δ : gain over LR_{PH} (pp). Shared weights for LR_{4S} . Earlier selection settings in supplement. CS: ConstructionSite.

Variant	Score	CUB	gRef	NLVR	CS	Avg.	Δ
LR_{PH}	r	65.86	59.37	87.33	21.24	58.45 (0.07)	–
3-state	$p_R - p_H$	71.27	60.13	86.34	21.77	59.88 (0.06)	+1.43
Gain	$\hat{\Delta}$	71.01	59.97	86.29	21.61	59.72 (0.04)	+1.27
LR_{4S}	$p_R - p_H$	71.30	60.10	86.43	21.69	59.88 (0.06)	+1.43
LR_{4S}	$\log(p_R/p_H)$	67.80	59.84	87.36	21.11	59.03 (0.09)	+0.58
LR_{4S}	r_{4S}	67.63	59.90	87.36	21.30	59.05 (0.10)	+0.60
LR_{4S}	s_α	71.37	60.66	87.36	21.63	60.25 (0.06)	+1.81

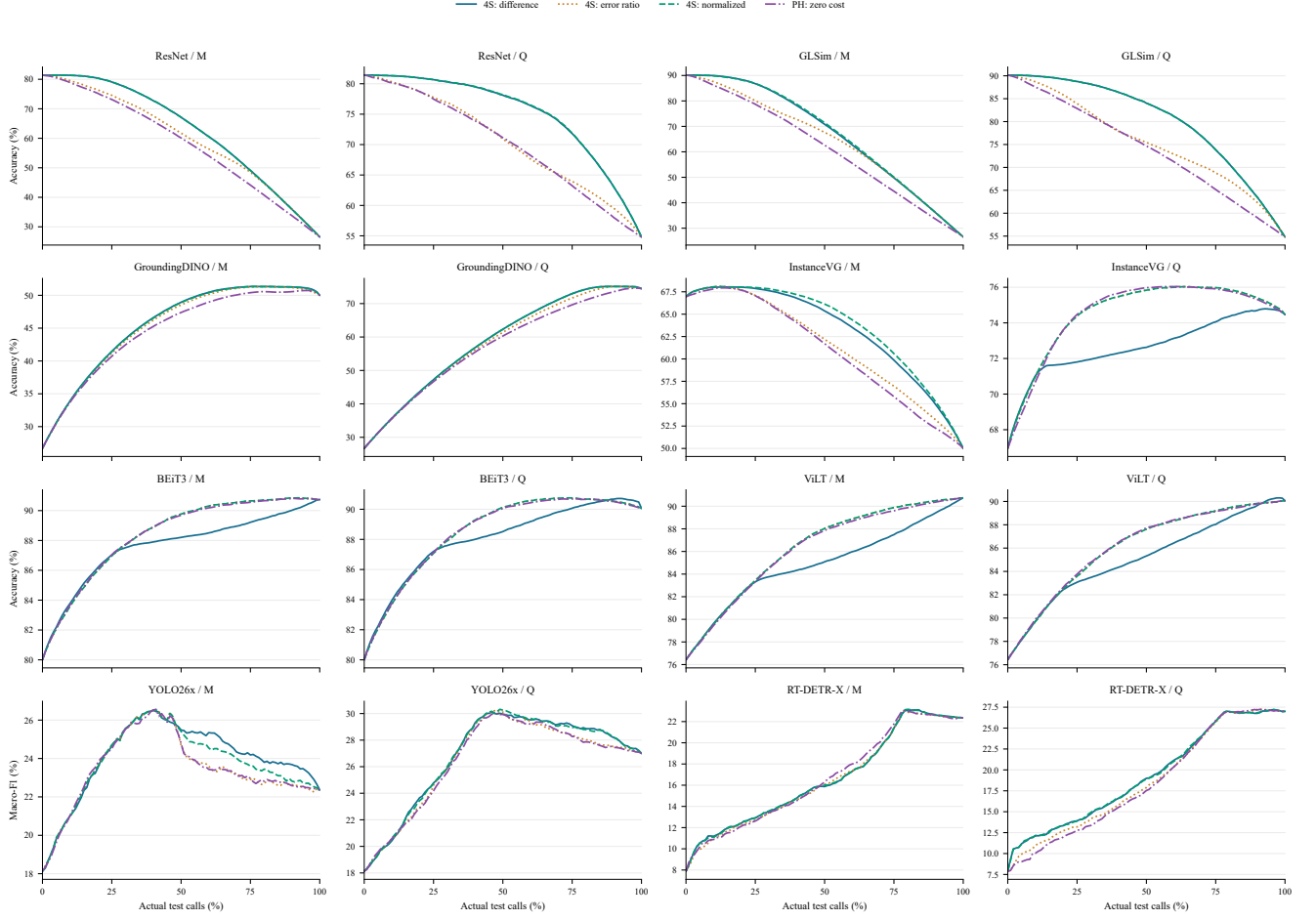


Figure 5: Earlier experiments: changing four-state scoring while holding weights fixed, with separately trained zero-cost post-hoc deferral. Both use the archived checkpoint-selection range. These are equal-call curves, not test-selected best-threshold envelopes.

The full-curve three-state versus four-state differences are below 0.1 points for each task. On NLVR2, changing the same four-state checkpoints from probability difference to error-ratio ranking improves the 25–100% curve mean from 87.82 to 89.09, close to post-hoc’s 89.03. On CUB, the ratios instead reduce performance. This supports a task-dependent scoring effect, not a claim that supervision never matters. The true conditional gain remains optimal for equal costs and additive accuracy; estimated probabilities need not preserve its ideal ranking. Construction macro-F1 is non-additive.

The earlier eighteen-setting architecture search, data-fraction studies and joint-training experiments retain their archived settings; they are not ablations of the final full-budget recipe. They are not used here to claim that each input branch is necessary. Similarly, the older objective table that mixed separately selected ratio checkpoints is superseded by the same-checkpoint scoring comparison above.

6 Choice of Deferral Baselines

Post-hoc deferral [1] learns when to replace an existing predictor’s answer, which matches the fixed specialist answers used here. Joint classification–deferral methods [6, 7], including OvA, instead learn class outputs together with a deferral output. Using their native class branch would change the answer when the VLM is not called; replacing that branch with a fixed specialist would change the original method. Moreover, a single class output does not directly represent a set of grounding boxes or multiple safety-event labels. These methods can be studied on classification subsets or with explicit adaptations, but are not interchangeable baselines for all four tasks under fixed answers. The event-averaged post-hoc loss used for construction is also an adaptation; we do not claim the original classification guarantees for macro-F1.

7 Cost Estimates

Figures use measured component costs for the same router architecture: specialist and router costs are paid on every input, with VLM cost added at the actual call rate. Confidence omits the router. Component measurements used 64 inputs at batch size 1 on an A800. These are serial component-based estimates, not newly measured integrated latency; the extra normalization arithmetic was not separately benchmarked. Costs at the selected thresholds have been recomputed from the new call rates.

7.1 Prompts and Generation Settings

The following templates come from the paired-prediction collection. Number words for 4 and 16 are displayed as digits here. The original prompt files are retained with the code. Placeholders are replaced by each sample's expression or statement. CUB supplies all 200 candidate classes in a single request; its full class list is supplied with the code. Both VLMs receive the same task input, but their templates may differ. All requests use temperature zero, with limits of 8 tokens for CUB and NLVR2, 512 for gRefCOCO and 768 for ConstructionSite. NLVR2 sends the left image followed by the right image, with these roles explicitly marked. Qwen disables thinking for CUB and NLVR2; MiniCPM constrains these answers to the class index or True/False. Localization and event requests use JSON schemas. Invalid answers are retained and counted as errors without repair or a second model call.

7.1.1 CUB-200-2011

MiniCPM-V-4.5

Identify the bird species in this image using the fixed CUB-200-2011 class list below. Choose the class whose visible
→ shape, bill, head pattern, wing pattern, body color, and tail best match the bird. Ignore the background and any
→ non-bird objects. Reply with exactly one class number from 1 to 200 and no other text.

{class_names}

Qwen3.8-27B-FP8

Identify the bird species in the image from the fixed CUB-200-2011 reference below. Each class is paired with three
→ visual traits learned only from official training images. Use the species name to establish the likely bird group,
→ then distinguish close candidates by the visible bill, crown and face markings, wing pattern, breast and belly
→ color, tail, and overall form. The traits are clues rather than absolute rules because pose, sex, age, lighting,
→ and occlusion can change what is visible. Choose the strongest overall match, ignore the background, and reply
→ with exactly one class number from 1 to 200 and no other text.

{class_traits}

7.1.2 gRefCOCO

MiniCPM-V-4.5

Your entire reply is one complete JSON line that begins with {"objects": and ends with the final }. A no-match answer
→ is exactly {"objects":[]}. A one-match answer looks like {"objects":[{"bbox_2d":[120,85,640,910],"score":0.92}]}.
→ Several matches use separate complete entries, for example
→ {"objects":[{"bbox_2d":[80,100,300,700],"score":0.93}, {"bbox_2d":[560,120,820,740],"score":0.91}]}. Every bbox_2d
→ contains all 4 xxyy numbers from 0 to 1000 with x1 smaller than x2 and y1 smaller than y2. Never end after only
→ one coordinate; finish every object, the objects list, and the final brace. Return no other text.

Locate all and only the visible objects that satisfy the entire referring expression below. Every stated type,
→ appearance, position, and spatial relation matters. Interpret left and right from the viewer's perspective, and
→ use front, back, nearest, farthest, and ordinal words to distinguish among similar objects. Some expressions
→ intentionally have no matching object, so use the no-match answer whenever the full expression is not visibly
→ satisfied instead of choosing the closest-looking object. Return one tight box for every distinct match without
→ merging objects into a group box.

Referring expression: {expression}

Qwen3.8-27B-FP8

Locate all and only the visible objects that satisfy the entire referring expression below. Every stated object type,
→ appearance, position, and spatial relation matters. Do not return the closest-looking object when part of the
→ expression is missing or contradicted. Some expressions intentionally have no matching object, so an empty list is
→ a normal answer whenever the full expression is not visibly satisfied.

The expression may identify zero, one, or multiple objects. When separate target descriptions are joined by words such
→ as "and", return a separate tight box for every described target rather than one box around the group. When "and"
→ only joins properties of one object, all those properties must hold for that one box. Do not merge neighboring
→ objects into one box.

Coordinates must be xxyy numbers normalized from 0 to 1000, and each box must have a confidence score from 0 to 1.

→ Reply with exactly one complete JSON object and no Markdown or explanation, using this form:
→ {"objects":[{"bbox_2d":[120,85,640,910],"score":0.92}]}. An answer with no match is exactly {"objects":[]}.

Referring expression: {expression}

7.1.3 NLVR2

MiniCPM-V-4.5

The first image is the left image and the second is the right image. Determine whether the sentence below is true for
→ this ordered pair. A sentence is True only when every part of it is supported by the images. Interpret counts and
→ comparisons exactly, distinguish claims about one image from claims about both images together, and handle words
→ such as all, at least, more, fewer, same, different, and no literally. Ignore details that the sentence does not
→ mention. Answer with exactly True or False and nothing else.

Sentence: {sentence}

Qwen3.8-27B-FP8

The first image is the left image and the second is the right image. Decide whether the sentence below is true about
→ the pair. Count each distinct matching object once in the exact image scope named by the sentence before
→ evaluating more, fewer, equal, at least, at most, or exactly. An identifiable partly visible object still counts,
→ but background patterns do not. Check the named colors, shapes, identities, negations, and spatial relations, and
→ require every stated clause to hold without adding unstated conditions. Reply with exactly True or False and no
→ other text.

Sentence: {sentence}

7.1.4 ConstructionSite

MiniCPM-V-4.5

Look at this one construction-site image and report all visible violations from the 4 named categories in one JSON
→ answer. Mark a category true when the image gives clear visual evidence. Do not invent details that are hidden,
→ but do not reject a plainly visible violation merely because the scene is not a close-up.

rule_1_ppe_violation: box every worker on foot who is visibly missing a hard hat or another clearly required item of
→ personal protection. A bare head is positive. Do not assume hidden footwear or clothing is missing, and do not use
→ the lack of a reflective vest in ordinary daylight by itself.

rule_2_fall_protection_violation: box every worker clearly about three metres or more above the lower level when the
→ exposed side has no effective edge protection and no safety harness is being used. Ground workers, low platforms,
→ dirt slopes, effective guardrails, and uncertain heights are negative.

rule_3_unprotected_edge_violation: box every people-accessible, continuous upper edge of a visibly deep excavation or
→ steep drop of about three metres or more when guardrails, fencing, and warning protection are absent. The image
→ must show both a substantial lower level and its hazardous upper edge; excavation activity or exposed soil alone
→ is not enough. Soil piles, excavation walls without an accessible upper edge, shallow trenches, gentle slopes,
→ ordinary uneven ground, uncertain distant cuts, and protected edges are negative. Box the hazardous upper edge and
→ only enough adjacent drop to identify it, never a worker or the whole image.

rule_4_excavator_proximity_violation: box a worker when the worker is clearly in the excavator's rear blind spot,
→ swing radius, boom or bucket path, or close enough beside the body or tracks to be reached by normal machine
→ movement. A worker merely visible in the same scene is not enough. The cab operator, a worker behind a barrier,
→ and a worker separated from the machine by a clearly safe gap are negative. Box only the endangered worker.

Use exactly this JSON shape:

```
{
  "events": {
    "rule_1_ppe_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    },
    "rule_2_fall_protection_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    },
    "rule_3_unprotected_edge_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    },
    "rule_4_excavator_proximity_violation": {
      "present": false,
      "reason": null,
      "boxes": []
    }
  }
}
```

Change a visible category to present=true, give it a short positive reason, and add one or more worker or edge boxes.
→ The word false must always keep reason=null and boxes=[]. Coordinates are integer image-relative [x1,y1,x2,y2]
→ values from 0 to 1000 with x1 < x2 and y1 < y2. Use at most 13, 6, 3, and 2 boxes for rules 1 through 4. Return
→ only the JSON object, with no Markdown or extra text.

Qwen3.8-27B-FP8

Look at this construction-site image and report every clearly visible violation from the 4 categories below in one
→ JSON answer. Most categories will usually be absent in a single image. A dangerous-looking construction scene is
→ not enough by itself; mark a category true only when its specific target and violation are visible.

rule_1_ppe_violation means a worker on foot is visibly missing required personal protection. A clearly bare head is a
→ violation. A worker who visibly wears a hard hat should normally be treated as compliant unless another required
→ item is unmistakably missing for the task shown; ordinary work clothes or the absence of a reflective vest in
→ ordinary daylight are not enough by themselves. Do not guess about small, distant, blurred, or hidden clothing,
→ footwear, eyes, or faces. Put a tight box around each violating worker's visible body, not around the missing
→ item.

rule_2_fall_protection_violation means a worker is clearly about three metres or more above a lower level, is exposed
→ because effective edge protection is absent, and is not using a safety harness. Workers on the ground, on a low
→ platform or dirt slope, behind effective guardrails, or at an uncertain height are not violations. Box each
→ exposed worker, not the structure or edge.

rule_3_unprotected_edge_violation means an accessible upper edge borders a visibly deep excavation or steep drop of
→ about three metres or more and lacks guardrails, fencing, or warning protection. The substantial height difference
→ and the hazardous upper edge must both be discernible in the image rather than inferred merely because excavation
→ work or exposed soil is present. Soil piles, excavation walls without an accessible upper edge, shallow trenches,
→ gentle slopes, ordinary uneven ground, distant cuts with uncertain depth, and protected edges are not violations.
→ Put one box around each continuous hazardous upper edge and only enough adjacent drop to show the hazard; do not
→ box workers or the whole image.

rule_4_excavator_proximity_violation means a visible worker is inside the normal reach of an excavator's rear
→ counterweight, rotating body, boom, or bucket. A still photograph need not show actual motion: count a worker who
→ is clearly within the machine's ordinary swing or travel zone, including directly beside its rear or tracks in the
→ active work area. Do not count the cab operator, a worker behind a barrier, a worker separated by a clearly safe
→ gap, or simple distant co-appearance. Box each endangered worker, not the excavator or a combined worker-machine
→ area.

Return exactly one JSON object whose "events" object contains "rule_1_ppe_violation",
→ "rule_2_fall_protection_violation", "rule_3_unprotected_edge_violation", and
→ "rule_4_excavator_proximity_violation" exactly once. A visible event is {"present":true,"reason":"short visible
→ evidence","boxes":[[x1,y1,x2,y2]]}; an absent or uncertain event is {"present":false,"reason":null,"boxes":[]}.
→ Coordinates are integer image-relative values from 0 to 1000 with $x_1 < x_2$ and $y_1 < y_2$. Return at most 13, 6, 3,
→ and 2 boxes for rules 1 through 4. Return no Markdown or text outside the JSON object.

7.2 Earlier Prompt Comparisons

Figures below reproduce earlier prompt comparisons on training subsets: 300 expressions for gRefCOCO and 400 examples each for CUB and NLVR2. Construction uses its recorded training selection. The source label Construction10K refers to ConstructionSite. These curves use the earlier task metrics (including gRefCOCO F1), so their absolute values are not directly comparable with the final routing table. They document prompt choice, not router ablations or new test-time tuning. The experiment ledger retains the input subset and metric for every point.

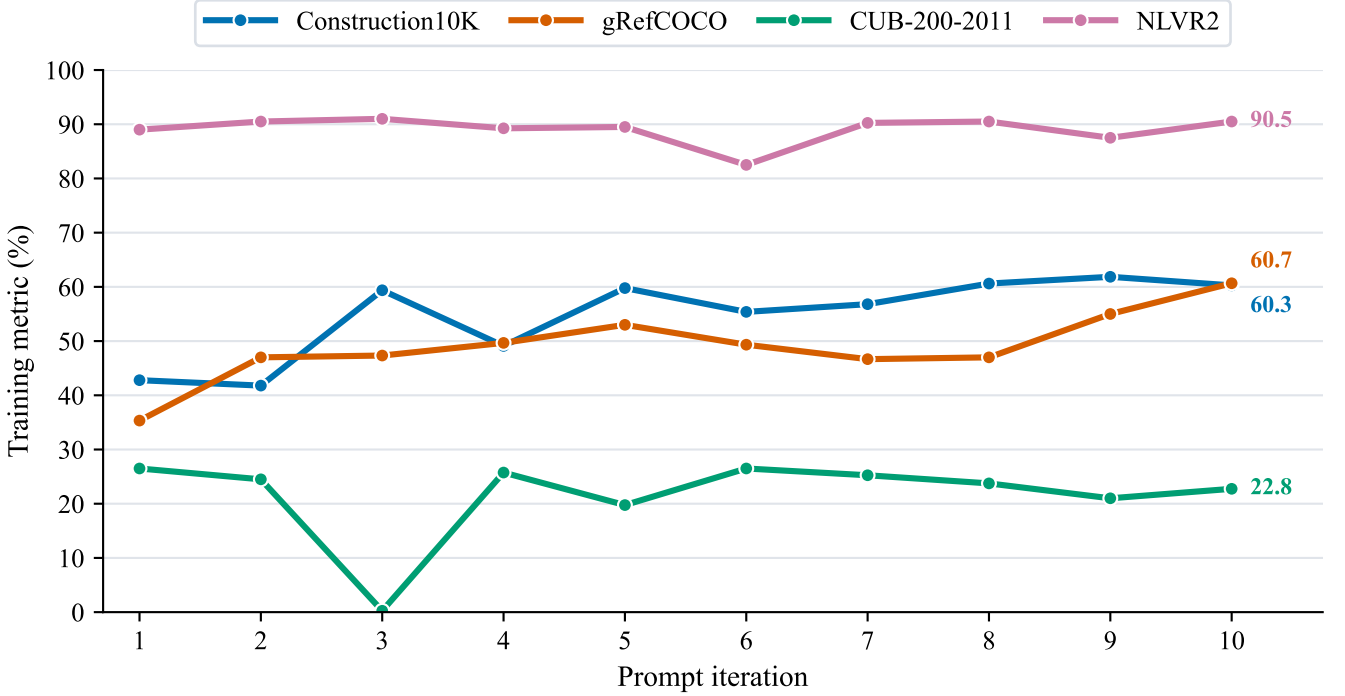


Figure 6: Earlier MiniCPM prompt comparisons from the recorded experiment ledger.

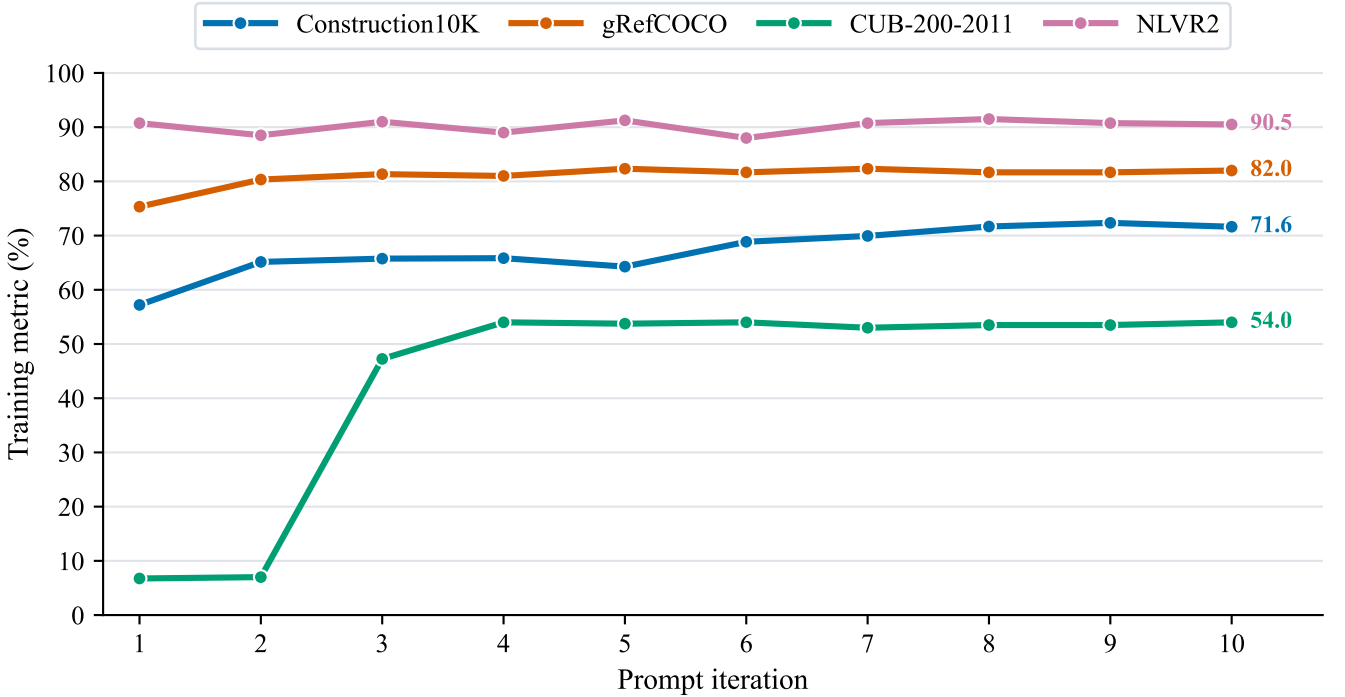


Figure 7: Earlier Qwen prompt comparisons from the same ledger.

References

- [1] Harikrishna Narasimhan, Wittawat Jitkrittum, Aditya K. Menon, Ankit Rawat, and Sanjiv Kumar, “Post-hoc estimators for learning to defer to an expert,” in *Proc. NeurIPS*, 2022, vol. 35, pp. 29292–29304.
- [2] C. Wah, S. Branson, P. Welinder, et al., “The Caltech-UCSD Birds-200-2011 dataset,” Tech. Rep. CNS-TR-2011-001, California Institute of Technology, 2011.
- [3] Chang Liu, Henghui Ding, and Xudong Jiang, “GRES: Generalized referring expression segmentation,” in *Proc. IEEE/CVF CVPR*, 2023, pp. 23592–23601.
- [4] Alane Suhr, Stephanie Zhou, Ally Zhang, et al., “A corpus for reasoning about natural language grounded in photographs,” in *Proc. ACL*, 2019, pp. 6418–6428.
- [5] Xuezheng Chen and Zhengbo Zou, “Are large pre-trained vision language models effective construction safety inspectors,” *Data-Centric Engineering*, vol. 7, pp. e11, 2026.
- [6] Hussein Mozannar and David Sontag, “Consistent estimators for learning to defer to an expert,” in *Proc. ICML*, 2020, pp. 7076–7087.
- [7] Rajeev Verma and Eric Nalisnick, “Calibrated learning to defer with one-vs-all classifiers,” in *Proc. ICML*, 2022, vol. 162, pp. 22184–22202.